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SUCCESSOR Ω — ANABASIS Ω

The Expansion Institution

From evidence capital to qualified productive worlds.

Own the mission. Price the proof.
Build only what the evidence earns.

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Prepared for author review · AI-assisted research edition

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Conceptual AI-generated imagery. Exact claims are controlled by the typeset text and evidence records.

Publication record and evidence contract

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The institutional proposition

Retain an inspectable method for deciding what is worth proving next. Learn useful world programs, compare complete alternatives, purchase consequential evidence, and preserve the option not to build.

Four kinds of statement. Source-defined architecture is attributed to the supplied papers. Retained results come from the delivered synthetic experiment. New derivations and numerical extensions are labelled as this publication's analyses. Field superiority, independent admission, operational authority and stellar infrastructure remain prospective.

The fictional owner, physical process, hidden states, observation channels and monetary quantities are authored scenario assumptions. Exact finite calculations establish consequences of those assumptions, not their truth. No return on a financial security, actual customer receipt or investment recommendation is represented. Product ownership and third-party rights remain governed by the applicable agreements and licences.

The generated art conveys the desired horizon. Its incidental text and decorative curves are not scientific diagrams or evidence. Technical figures are generated from included code. A checksum checks bytes; it neither authenticates a publisher nor establishes independent proof.

This edition does not retroactively alter the original trial, transform a conventional-comparator tie into proprietary Alpha, or claim unrestricted scientific theory discovery. The new physical derivations do not imply that a corporation obeys a mechanical least-action law.

Abstract

A productive institution must decide not only how to operate its assets but which next capability deserves existence. We develop *SUCCESSOR Ω — ANABASIS Ω* as a source-grounded research formulation of that expansion problem. The supplied GoalOS + SEIZE and Mission Gym constitutions are instantiated through a retained finite, partially observed customer study: eight latent states, five assays, twelve terminal designs and a bounded proof-capital budget. Four local neural guides and twelve symbolic generations are produced by the v10.2 scientific engine. All four G2 candidates pass a nominal synthetic predictive protocol and fail a later shift; three G3 candidates recover local validity. The contingent policy begins with a CA$ 450,000 assay, produces CA$ 104.664 million expected project net present value after assays, and chooses no-build in 47.86% of modeled outcomes. It exceeds one-step research by CA$ 50.791 million but ties an informed conventional predictor using the same complete planner. A CA$ 12 million programme allowance therefore does not earn an exclusive model premium.

The decisive error lies in evidence dependence: two positive, shared-calibration assays imply 14.66% unsuitable-recovery probability in the declared latent-state model, versus 8.80% when dependence is omitted. This publication freshly recomputes the original study and adds a fixed-policy channel sensitivity analysis. The frozen 10% loss gate is breached after a 7.532% interpolation toward the authored adverse pilot channel; that interpolation is not an empirically estimated contamination rate. A separate linear programme exposes inspection and then electricity as expansion bottlenecks. Conditional propositions develop Bayesian control, the limited transfer of information-ordering results to posterior risk gates, total-variation certificates, truthful scoring and information-acquisition incentives, enforceable audits, bargaining with portable outside options, resource shadow prices, material renewal, separation exergy, thermal passivity and finite-speed control.

The resulting vision is a sequence of qualified productive institutions: knowledge changes decisions; accepted decisions change capacity; collected surplus funds the next observation and successor. The construction remains conditional. No independent Specialist ASI admission, customer Alpha, unrestricted theory discovery or production authorization is asserted.

Keywords: neural-symbolic world programs; Mission Gym; proof capital; correlated evidence; real options; mechanism design; exergy; renewal; mission sovereignty.

The owner's one-page decision

Buy the right observation before buying the next factory

The strongest use is a high-consequence, measurable decision with several credible alternatives and a bounded, decision-changing experiment. The owner retains the mission machinery even when a simpler predictor is the better supplier.

Decision-relevant finding	Result
Expected project NPV after assays	CA$ 104.664 M
Contrast over one-step research	CA$ 50.791 M
Expected assay expenditure	CA$ 2.859 M
Probability of no-build	47.86%
Maximum nominal reached-leaf severe-loss risk	7.72%
Same-planner informed Beta	Tie
Net comparison after assumed programme cost vs informed Beta	CA$ −12 M
Real customer Alpha credited	CA$ 0

Do now. Preserve the contingent policy, the negative result against informed Beta, the assay-bias explanation and the stopped branches. Commission a measurement plan that challenges the pilot's conditional validity. Price formation, independent review, integration and unwind before requesting a model premium.

Do not do. Add scenario differences together, call projected NPV reinvestable cash, treat a local test as independent proof, or transfer a descendant's authority by name.

Reading routes. Owners: Sections 1–7, 20–23 and the practical guide. Scientists: Sections 3–6 and 8–19. Engineers and verifiers: Section 22, the source map, code, fixtures and validation record. The exact canonical progression is retained: Successor Manifold → Mission Advantage Gradient → SEIZE → Bounded AGI Jobs → Specialist ASI → Mission-Sovereign Successor. The final two are earned states, not completed stages in this research package.[1, 2, 3]

Contents

I

The institution of expansion

A consequential question before a capital commitment.

CONCEPTUAL ART · NOT A MEASUREMENT OR DEPLOYMENT

1 From operating continuity to the next capability

AEON studied continuity of an existing estate. ANABASIS studies a different question: what productive capability should the estate add, what observation could change that decision, and what should remain owned after a preferred model loses? The object is neither a permanently dominant predictor nor an unrestricted agent. It is a succession-capable institution that can retain its mission, evidence and refusal conditions while its computational suppliers change.[3]

Dr. Soraya Delcourt is the fictional principal of a CA$ 620 billion industrial estate with CA$ 18 billion in group liquidity. These values establish the story's setting; they are not assets of the commissioning user and do not prove project affordability. Her ring-fenced mission permits at most CA$ 750 million of construction and CA$ 6.5 million of assays. The alternatives are resource recovery, precision fabrication, an integrated line, or no new asset. Each construction choice has a two-year commissioning delay and a twelve-year pre-tax horizon at an assumed 8% discount rate.[4]

Her rational ambition is not maximal expenditure. It is to increase the set of useful, supportable future actions. A small diagnostic can be worth more than an impressive integrated facility when it prevents a wrong irreversible commitment. Conversely, a negative initial estimate can justify investigation when an inexpensive observation exposes valuable contingent choices.

The owner's mandate

Define the decision before the factory. Purchase evidence before promotion. Count accepted material and external sales. Keep every reason to stop. Reinvest only value whose state permits it.

2 The source architecture and the implemented boundary

2.1 Three objects that must remain distinct

The Gym paper's triad is operationally decisive. The *Mission Gym* represents the mission world and its rules. *Specialist ASI* is an earned system-level comparative designation. The *Mission-Sovereign Successor* is that proven capability after separate accountable admission. The table on PDF page 16 of the supplied Gym paper gives each object a different question; its architecture diagram on PDF page 23 places models inside a larger institution.[2]

In ANABASIS, the native v10.2 scientific engine supplies numerical-expression search, fitting, local neural-surrogate training and experiment proposals. Customer adapters supply the process account, hidden-state evidence model, contingent planner, Gym episodes and economic interpretation. A customer's correctly authored likelihood table is not silently credited to autonomous discovery by the native engine.

2.2 The canonical progression, faithfully applied

The Navigator defines SEIZE as *Surface, Evaluate, Instantiate, Zero in, Elevate*. It freezes the mission, relevant alternatives, hard gates and proof budget; it then submits one exact challenger for examination. SEIZE issues a promotion request, not operating authority. Its five decision functions—Underwrite, Execute, Accept, Admit, Allocate—are separate from the corporate D0–D4 phase gates and the A0–A4 authority ladder.[1, 3]

The supplied Gym diagram on PDF page 72 organizes the full cycle into four phases: sense and underwrite; form and freeze; prove and admit; operate, learn and succeed. Here, the first two phases have local research implementations. Protected independent custody, field proof, accountable admission and real operation remain obligations. Files named `QUALIFICATION.json` or `BOUNDED_JOBS.json` document this boundary; they do not issue credentials.

2.3 What the institution retains

Subject to actual rights, the durable layer contains the constitution, authorized observations, model programs, tool contracts, baselines, tests, rejected alternatives, exposure history, evidence references and recovery instructions. Retaining that layer enables technical substitution. It does not automatically transfer third-party weights, legal rights or a predecessor's proof.[3]

3 The mission as a finite belief-state problem

Let the hidden world be $w = (r, f, b) \in \{0, 1\}^3$: recovery suitability, fabrication suitability and persistent calibration bias. The study therefore has eight states. Five one-use evidence actions return binary observations. Twelve terminal choices comprise retain, five recovery intensities, five integrated intensities and one fabrication choice.[4]

A history $h = ((e_1, y_1), \ldots, (e_k, y_k))$ produces belief

$$p_h(w) = \frac{p_0(w) \prod_{(e,y)\in h} L_e(y \mid w)}{\sum_v p_0(v) \prod_{(e,y)\in h} L_e(y \mid v)}. \tag{1}$$

The product is valid because observations are conditionally independent *given the full hidden state*, including persistent bias. Removing that state and multiplying marginal assay probabilities changes the model.

Let $U(d, w)$ be terminal project NPV, before assays and the separate programme allowance. Let c_e be an assay's complete declared scenario cost. A terminal design is feasible at history h when it satisfies construction limits and

$$\sum_w p_h(w)\mathbf{1}\{U(d, w) < -100\} \le 0.10, \tag{2}$$

with monetary units in millions of Canadian dollars. This is a specific posterior project-loss constraint, not a frequentist return guarantee or a complete tail-risk model. The original loss event excludes assay and programme costs; a separately named diagnostic later checks an all-in alternative without rewriting history.

The admissible experiment set $\mathcal{E}(h)$ excludes repeated tests, unaffordable tests and histories already using three tests. With $\mathcal{D}(h)$ denoting feasible designs,

$$\begin{aligned} V(h) = \max\Big\{ &\max_{d\in\mathcal{D}(h)} \sum_w p_h(w)U(d, w), \\ &\max_{e\in\mathcal{E}(h)} \Big[-c_e + \sum_y P(y \mid h, e)V(h \parallel (e, y))\Big]\Big\}. \end{aligned} \tag{3}$$

Already-paid assay fees are sunk in $V(h)$ and counted once in the path ledger. The finite formulation specializes belief-state planning to bounded proof capital.[5]

Proposition 1 (Optimality inside the declared finite mission)**.** *If the hidden-state prior, observation kernels, terminal utilities, history-dependent constraints and finite test catalogue are fixed, recursion* (3) *maximizes expected terminal utility less future assay costs over admissible deterministic contingent plans.*

Proof. At histories with no permissible experiment, only feasible terminal choices remain. Assume the continuation values are optimal for every shorter remaining horizon. Each admissible next test incurs its fee once and branches into exactly those continuations, weighted by its observation probabilities. Maximizing over that set and stopping completes backward induction. Randomization cannot improve a maximum of expectations over a finite menu. The result is about the specified world, not unknown reality. □

4 Learned process programs, not decorative equations

The physical-response adapter models recovered mass fraction as a function of normalized intensity u and normalized contamination z. The authored nominal mechanism is

$$y(u, z) = 0.50 + 0.34u - 0.09u^2 - 0.23z - 0.05uz, \tag{4}$$

with uniform synthetic measurement noise of amplitude 0.0002. Knowing this generating mechanism is a property of the study's author, not information granted to the learner. Development and challenge records have different identities, but all remain under common author custody.[4]

The native engine performs 8,935 candidate fits across four cohorts, trains four twelve-hidden-unit neural guides for 1,000 steps each, and retains twelve generation artifacts. Removing the neural contribution changes the proposed calibration probe in all four cohorts. This establishes computational participation, not global optimality of active learning.

The first G2 program is approximately

$$\begin{aligned}\widehat{y}_2(u, z) = 0.401267 - 0.229983z + 0.378340u - 0.050054zu \\ + 0.136798 \cos u - 0.038130e^u.\end{aligned} \tag{5}$$

It is used in the terminal material and economic calculations. Its trigonometric and exponential terms approximate the bounded response; they do not identify a unique physical mechanism. All four G2 candidates pass the local nominal prediction protocol. A later authored shift adds $-0.065z^2$ to the generating relationship: all four G2 candidates fail, while three G3 candidates pass the changed protocol. One unsuccessful descendant is preserved.

The useful unit is a frozen executable hypothesis with inputs, units, domain, evidence, loss and falsifier. Program synthesis research supports this general direction, but neither executability nor a successful fit establishes causal truth or field competence.[6, 3]

5 What the contingent institution actually decides

The central policy first purchases recovery coupon B for CA$ 450,000. A positive result leads to the CA$ 350,000 calibration blank. A negative blank supports the recovery choice within the declared model; a positive blank triggers the more costly joint pilot. A negative first coupon leads to coupon A; two negatives stop, and conflicting positives require the pilot. All reached branches, including no-build branches, are included in the denominator.[4]

Policy	Expected NPV	Assay spend
	CA$ million	
No research and no-build	0.000	0.000
One-step research	53.872	4.500
Information gained per dollar	96.751	2.676
Dependent contingent plan	104.664	2.859
Informed OLS with the same planner	104.664	2.859

The contingent plan chooses no-build with probability 47.856%. It exceeds the one-step strategy by CA$ 50.791 million, but that difference is already inside the project's CA$ 104.664 million value. It is not an additional revenue stream. The central configuration was selected during scenario design to expose sequential option value and correlated evidence; it is not representative sampling of industrial projects.

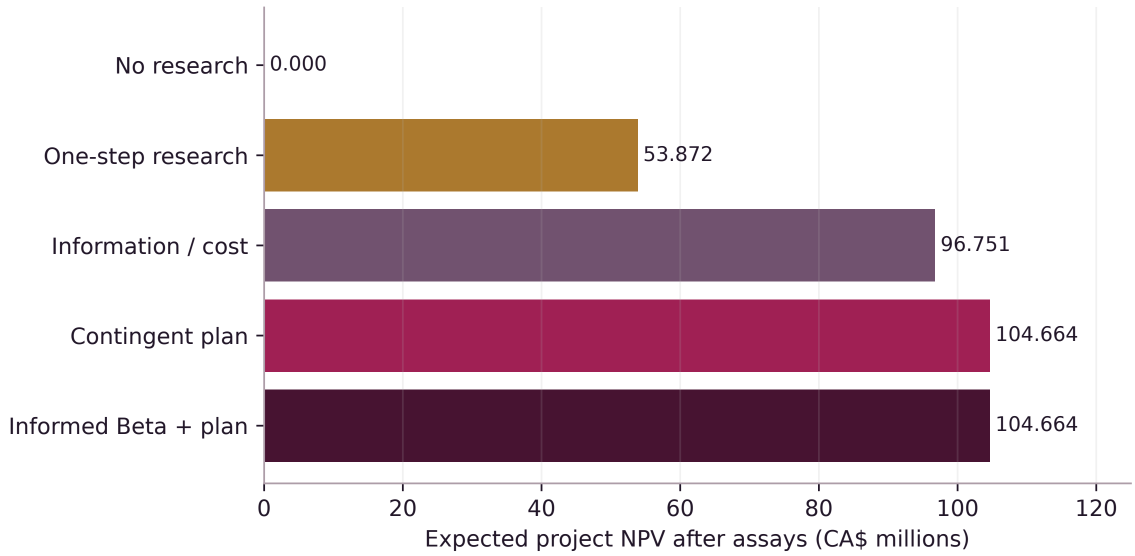

Figure 1: Retained synthetic comparison, freshly recomputed. Project NPV includes assay costs, but not the separate CA$ 12 million programme allowance. Informed Beta ties the complete method.

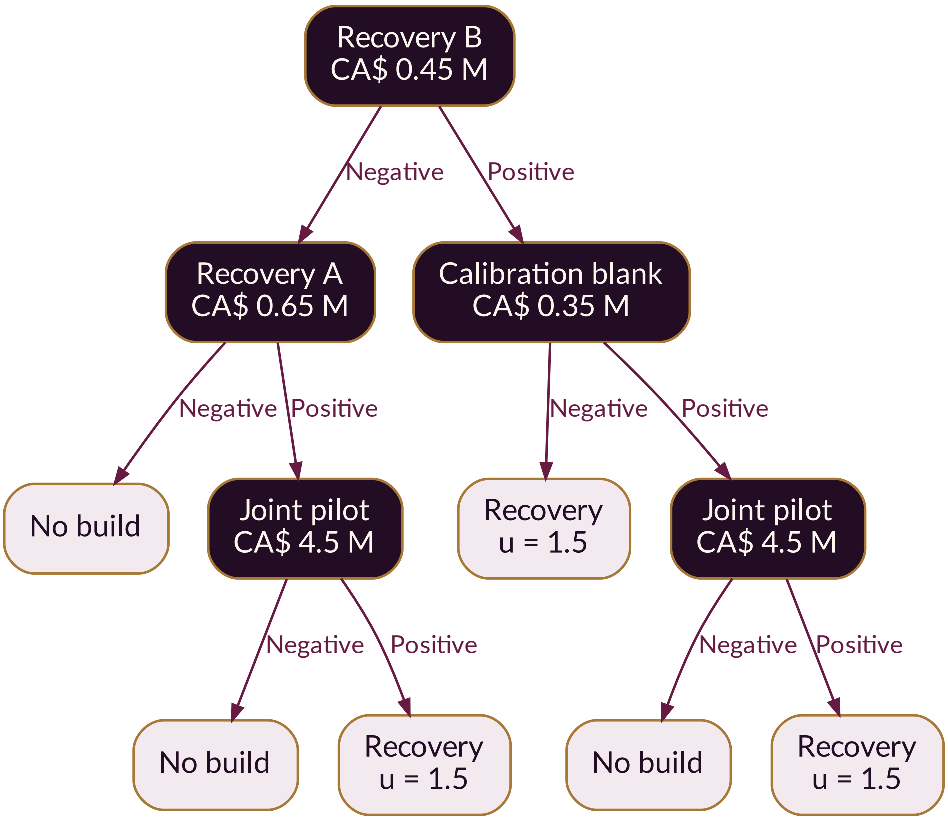

Figure 2: Exact retained contingent policy. Labels represent hypothetical assays and model-based terminal choices, not permission to purchase or construct.

Proposition 2 (Leaf-wise constraints imply an unconditional bound)**.** *If every reached investment leaf has posterior severe-loss probability at most ρ, no-build has zero project severe-loss probability, and branch probabilities are correctly specified, then*

$$P(\textit{severe loss}) \leq \rho P(\textit{invest}). \tag{6}$$

Proof. Partition outcomes by terminal history and apply the law of total probability. Each investment term is at most ρ times its history probability, and no-build contributes zero. In the retained case, $0.037303 \leq 0.1(1-0.478561)$. This conclusion inherits the likelihood and loss-definition assumptions; it does not survive arbitrary misspecification. □

6 The evidence channel is itself a hypothesis

Two positive recovery coupons do not constitute two independent witnesses. Their shared bias can explain both. After those positives, unsuitable-recovery probability is 14.658% in the complete latent model and 8.798% in the independence-only approximation. The latter crosses below the frozen 10% gate and changes the investment recommendation.

The old Gym's frozen plan has an attractive CA$ 117.120 million expected value under nominal physical evaluation, yet 58.0804% of its reached-leaf probability mass is on branches violating the true nominal risk constraint. This is not a 58% probability of project loss. It is the probability of arriving at an inadmissible decision branch. Higher mean utility cannot compensate for that failure.[4]

6.1 New publication-side channel boundary

We retain the nominal dependent plan without retraining or re-optimizing it. Its observation kernel is challenged along

$$L_\lambda = (1-\lambda)L_0 + \lambda L_1, \qquad 0 \leq \lambda \leq 1, \tag{7}$$

where L_1 is the original adverse channel: when the hidden calibration-bias state is active, the joint pilot's positive probability becomes 0.84. The model's original pilot probabilities otherwise depend on recovery and fabrication suitability.

An executed 101-point scan and scalar root solve locate the first loss-gate crossing at $\lambda = 0.0753216368$. The maximum reached-leaf severe-loss probability increases from 7.716% at $\lambda = 0$ to 29.275% at $\lambda = 1$. *The 7.532% number is a mixture coordinate between two authored kernels, not an observed percentage of contaminated physical tests.*

The extension also recomputes a distinctly named loss event including assays and the CA$ 12 million programme allowance. The maximum reached-leaf probability happens to remain 7.716% in this discrete central case. Equal results here do not make the definitions interchangeable; a different utility or threshold can change which states count as losses.

7 Spread, Alpha and the informed alternative

The source architecture defines residual mission advantage relative to the strongest relevant baseline after the complete denominator is counted. The Gym version explicitly adds a Gym-to-reality reserve. These are governance definitions, not claims that local expected NPV already constitutes independently proven Alpha.[1, 2, 3]

Here, an informed ordinary-least-squares predictor receives the same observations, explicit mechanism features and the same planner. It chooses the same nominal decisions. This strong diagnostic comparator is not an authenticated current vendor quotation or an exhaustive survey of procurable alternatives.

A separate authored CA$ 12 million programme allowance covers formation, integration, proof preparation,

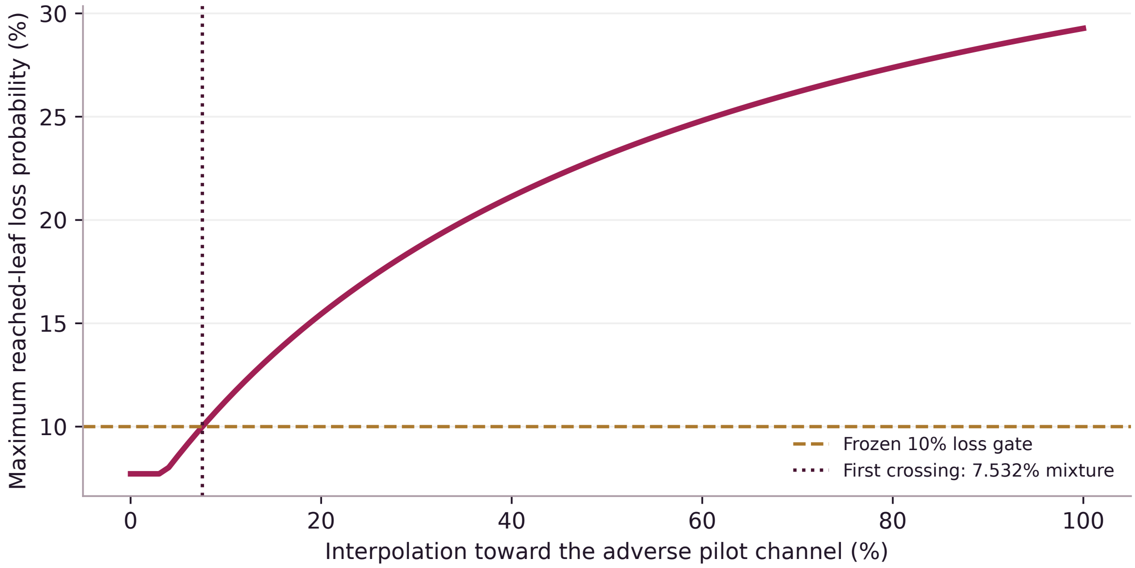

Figure 3: A fixed policy's admissibility can fail well before the full adverse channel is reached. No policy is re-optimized along this curve. The 10% gate and kernels remain explicitly scenario-defined.

human review, maintenance and unwind. After it, the complete candidate's project comparison is CA$ 92.664 million over no-build, CA$ 38.791 million over one-step research, CA$ –4.087 million against information-per-cost, and CA$ –12 million against informed OLS plus complete planning. These contrasts are mutually comparative, not additive.

The most useful ownership decision

Keep the mission, observation contracts, failure graph, world-program interface and evaluation method. Use the least-burden suitable predictor. Seek a premium only when a newly supported difference in evidence, process, operations or complete institutional cost survives the strongest fair comparison.

II

The mathematics of proof capital

Information has value when it changes an admissible decision.

CONCEPTUAL ART · NOT A MEASUREMENT OR DEPLOYMENT

8 Information ordering without overstating its reach

For a fixed feasible action set D and posterior p, define $v(p) = \max_{d\in D} \mathbb{E}_p U(d, w)$. Being a maximum of linear functions, v is convex. A more informative experiment can improve the attainable decision value when the decision maker is free to ignore information. This is the operational content relevant here from the comparison-of-experiments tradition.[7]

Proposition 3 (Priced evidence and the no-information option)**.** *Under a fixed action set and Bayesian coherence,*

$$\mathbb{E}_Y v(p_Y) - v(p_0) \geq 0. \tag{8}$$

A priced experiment is worthwhile only if this increase exceeds its total cost and any separately constituted constraints are satisfied.

Proof. The posterior is a martingale, $\mathbb{E}p_Y = p_0$. Jensen's inequality gives the result; alternatively, choosing the prior-optimal action for every observation reproduces the prior value. Subtracting a positive fee removes the nonnegativity guarantee for net value. □

ANABASIS uses mandatory posterior loss gates, so its feasible terminal menu changes with evidence. One cannot automatically import a fixed-action Blackwell theorem into that setting. An organization may not ethically or constitutionally ignore newly observed evidence merely to preserve yesterday's permission. The relevant value must be calculated in the actual constrained history tree.

A low-cost diagnostic can reveal much entropy without changing any permissible action. Conversely, a rare observation can have high decision value because it opens or closes a major investment. Information gained, model fit and mission utility are three different quantities.

9 What robustness certificates can honestly establish

Proposition 4 (Posterior event and decision sensitivity)**.** *Let p, q be probability distributions on a finite world, with $d_{\text{TV}}(p, q) = \frac{1}{2}\sum_w |p(w) - q(w)| \leq \delta$. For any event A, $|p(A) - q(A)| \leq \delta$. If each decision payoff has range width at most M, an optimizer under q has regret under p at most $2M\delta$, provided the same feasible decision set is used.*

Proof. The event inequality is the variational characterization of total variation. For a payoff in an interval of width M, shifting its lower endpoint to zero gives expectation error at most $M\delta$. Add and subtract the two q-expectations when comparing the p-optimal and q-optimal decisions; the q-optimality term is nonpositive, leaving two such errors. Different feasibility sets invalidate the last step. □

A conditional loss estimate below $\rho - \delta$ is robust to a *known* posterior-TV radius δ. Merely writing a radius in a manifest does not establish that reality lies inside it. Near an impossible observation, normalization can amplify small likelihood errors: if unnormalized masses differ by ϵ in ℓ_1, normalized TV is at most $\epsilon/\min(Z_p, Z_q)$ for positive evidential masses Z_p, Z_q. Rare histories can therefore need disproportionate calibration effort.

For a fixed policy over T evidence steps, a coupling argument bounds path-TV by prior-TV plus the sum of uniform stepwise kernel-TV errors, capped at one. This gives a finite-horizon diagnostic, not immunity to unmodeled latent states.

9.1 Robust recursion is a different constitution

One may replace each expectation by a worst-case value over an ambiguity set. A stagewise max–min recursion generally permits nature to change kernels at later histories. A single unknown, persistent biased instrument instead imposes cross-history coupling. Dynamic consistency of recursive multiple-prior models depends on structure such as rectangularity.[8]

This publication does not rename a stress grid as a fully robust optimizer. It reports frozen-policy sensitivity. A future robust mission must state whether uncertainty is one fixed world, a prior distribution over worlds, or an adversary choosing conditionals at each stage.

10 Truthful forecasting is not enough: pay for information

In the institutional game, the producer prefers promotion, the verifier may prefer low effort, and a provider may prefer dependence. Role separation changes opportunities but does not itself prove that incentives align. The supplied papers require separate producer, verifier and admission functions.[1, 2]

For a binary outcome Y and report q, consider a scaled Brier reward

$$S(q, Y) = s\big[1 - (q - Y)^2\big]. \tag{9}$$

Strict propriety addresses reporting conditional on beliefs. It does not by itself compel a verifier to acquire a better signal.[9]

Proposition 5 (Truthful reporting and costly signal acquisition)**.** *For subjective probability p, truthful reporting maximizes expected Brier reward and a report q loses $s(p-q)^2$. If an optional signal produces posterior p_Z at cost c, its expected reward improvement is $s\,\mathrm{Var}(p_Z)$; acquisition is privately worthwhile when this exceeds c.*

Proof. Expanding $\mathbb{E}(q - Y)^2$ gives $p(1-p) + (q-p)^2$. For an informative signal, total variance yields $p_0(1-p_0) - \mathbb{E}[p_Z(1-p_Z)] = \mathrm{Var}(p_Z)$. Multiply by s and subtract the signal cost. □

In an authored example, prior 0.5 and a symmetric 90%-accurate signal produce posterior values 0.9 and 0.1, hence variance 0.16. At $s = CA\$\,2,000$, the information reward is CA\$ 320; an effort cost of CA\$ 240 leaves CA\$ 80. This is a risk-neutral incentive calculation, not a proposed fee quote. Real outcome verification, conflicts, risk aversion, delayed labels and contractual enforceability require separate treatment. Rewards must not be contingent on issuing a pass.

11 Inspection incentives and enforceable sanctions

Let a producer gain b from misreporting. An audit occurs with probability a; detection probability is d under deception and false-alarm probability f under honest behaviour. A penalty K is enforceable with probability e. Honest reporting is not costless when audits can be wrong.

Proposition 6 (Audit deterrence with false positives)**.** *Under these risk-neutral assumptions, honesty weakly dominates deception if*

$$ae(d - f)K \geq b. \tag{10}$$

If $b/[e(d-f)K] > 1$, no feasible audit probability alone establishes deterrence.

Proof. Honest expected penalty is $aefK$. Deception adds benefit b and expected penalty $aedK$. Subtracting honest from deceptive payoff gives $b - ae(d-f)K$. Nonpositivity gives the condition. □

For $b = CA\$\,18,000$, $K = CA\$\,250,000$, $d = 0.92$, $f = 0.03$ and $e = 0.8$, the minimum audit probability is 10.112%. At $e = 0.05$, the required probability becomes 161.798%, which is impossible. The institution must change the incentive exposure, evidence custody, participation contract or action scope; it cannot fix limited enforceability by demanding “more vigilance.”

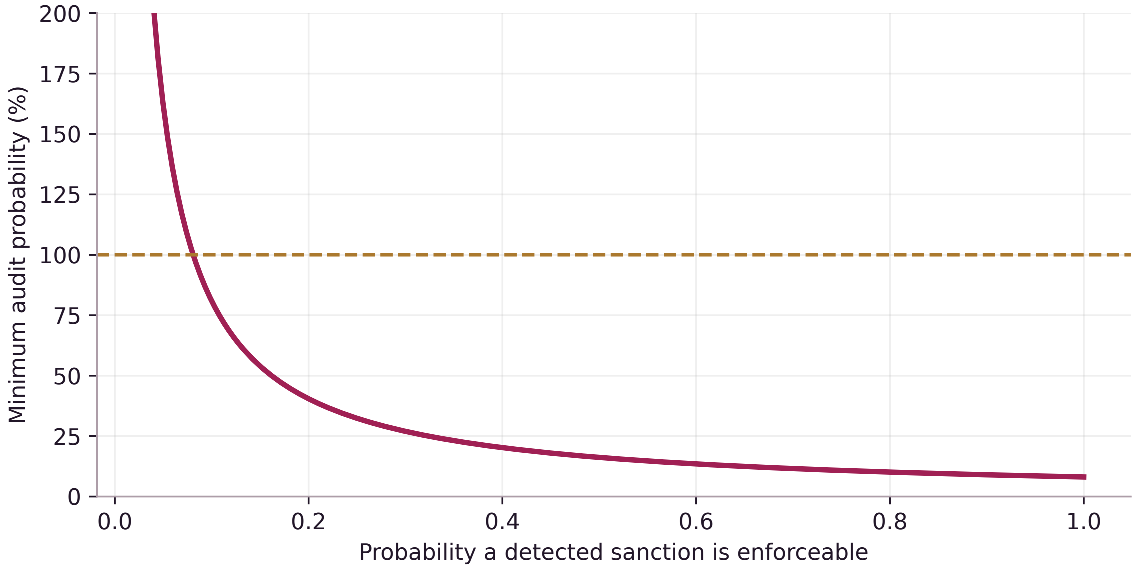

Figure 4: A conditional deterrence frontier. The curve above 100% identifies assumptions under which audit frequency alone cannot implement the desired reporting incentive.

12 Portability and bargaining over retained knowledge

The source-defined sovereignty premium concerns mission control and usable outside options, not ownership of every underlying model.[1, 2, 3] A stylized bargaining calculation makes the mechanism precise.

Proposition 7 (A portable outside option changes surplus capture)**.** *Let cooperative gross value be S, disagreement payoffs be d_P, d_V, and principal bargaining weight be $\theta \in (0,1)$. For $S > d_P + d_V$, the weighted Nash solution gives*

$$U_P = d_P + \theta(S - d_P - d_V). \tag{11}$$

Holding other terms fixed, an increase Δd_P raises principal payoff by $(1-\theta)\Delta d_P$, before portability cost.

Proof. Maximize $(U_P - d_P)^\theta (S - U_P - d_V)^{1-\theta}$ over feasible surplus divisions. The first-order condition yields the expression. Differentiating at fixed total surplus gives $1-\theta$. The surplus must remain positive, and changing technical capability can also change S. □

The formula specializes a bargaining model, not an empirical estimate of this product's commercial premium.[10] In practice, a credible outside option requires rights-cleared exports, working interfaces, equivalent tests and a restoration path. A folder of generated files may be portable storage without being portable mission capability.

13 Bounded search, information geometry and claim budgets

The supplied architecture uses manifold language as a disciplined design analogy. In discrete architecture coordinates, a gradient can be a finite difference or search signal. For a differentiable policy family, the Fisher matrix measures local distributional sensitivity. When it is singular, an ordinary inverse is not defined: one must restrict to identifiable coordinates, use a declared generalized inverse, or regularize $G = F + \lambda I$ with $\lambda > 0$.[2]

A related variational identity is useful without confusing computational temperature with kelvin.

Proposition 8 (Gibbs allocation among fixed candidates)**.** *For finite scores Q_i, $\tau > 0$ and distributions p, define $J(p) = \sum_i p_i Q_i + \tau H(p)$. With $g_i = e^{Q_i/\tau} / \sum_j e^{Q_j/\tau}$,*

$$\tau \log \sum_i e^{Q_i/\tau} - J(p) = \tau D_{\mathrm{KL}}(p \| g) \geq 0. \tag{12}$$

Proof. Substitute $\log g_i = Q_i/\tau - \log \sum_j e^{Q_j/\tau}$ into the definition of relative entropy. The identity follows term by term. Here τ has score units, not thermodynamic temperature. □

Repeatedly forming successors also creates repeated opportunities for false claims. If generation g has a valid conditional false-promotion probability at most α_g, then $P(\text{any false promotion}) \leq \sum_g \alpha_g$, without requiring independence. One schedule is $\alpha_g = 6\alpha/(\pi^2 g^2)$. It allocates, but does not create, valid evidence. Adaptive stopping within a generation requires an appropriate time-uniform method or a separate checkpoint error budget.[11]

A local registry, process boundary or hash does not establish independent custody. The current publication makes no confidence-backed customer-promotion decision.

The physical price of expansion

A surplus cannot repeal conservation, throughput or time.

CONCEPTUAL ART · NOT A MEASUREMENT OR DEPLOYMENT

14 The material account and the next bottleneck

For feed F, recovery yield y, fabrication transfer T and fabrication yield η_f, the retained adapter uses

$$R = Fy, \qquad W_R = F - R, \tag{13}$$
$$G = \eta_f(T + I), \qquad W_F = (T + I) - G, \tag{14}$$
$$S_R = R - T, \qquad F + I = W_R + W_F + S_R + G. \tag{15}$$

T is internal transfer, I is purchased fabrication input, S_R is externally sold recovered material and G is finished external output. Only external sales enter revenue. Electricity, fixed costs, disposal and initial capital are separately charged. The equations are a coarse synthetic account, not a process-engineering digital twin.[4]

In the separate throughput example, 135,000 tonnes of recovered feed supports at most 110,000 tonnes of fabrication input at 92% yield. Finished output could reach 101,200 tonnes, but an 80,000-tonne inspection ceiling binds. Buying more recovery feed does not increase accepted output.

This publication adds an electricity constraint to make the next binding resource inspectable. With variables (f, q) for fabrication input and accepted output,

$$\max q \quad \text{s.t.} \quad f \leq 135000,\ f \leq 110000,\ q \leq .92f,\ q \leq I_c,\ .8f + .15q \leq E_c,\ f, q \geq 0. \tag{16}$$

Energy coefficients are authored MWh per tonne. At $E_c = 130000$ MWh, raising I_c from 80,000 to 101,200 tonnes raises accepted output accordingly. At $E_c = 85000$ MWh, output is limited to 83,368.870 tonnes even with expanded inspection. No additional project savings are inferred.

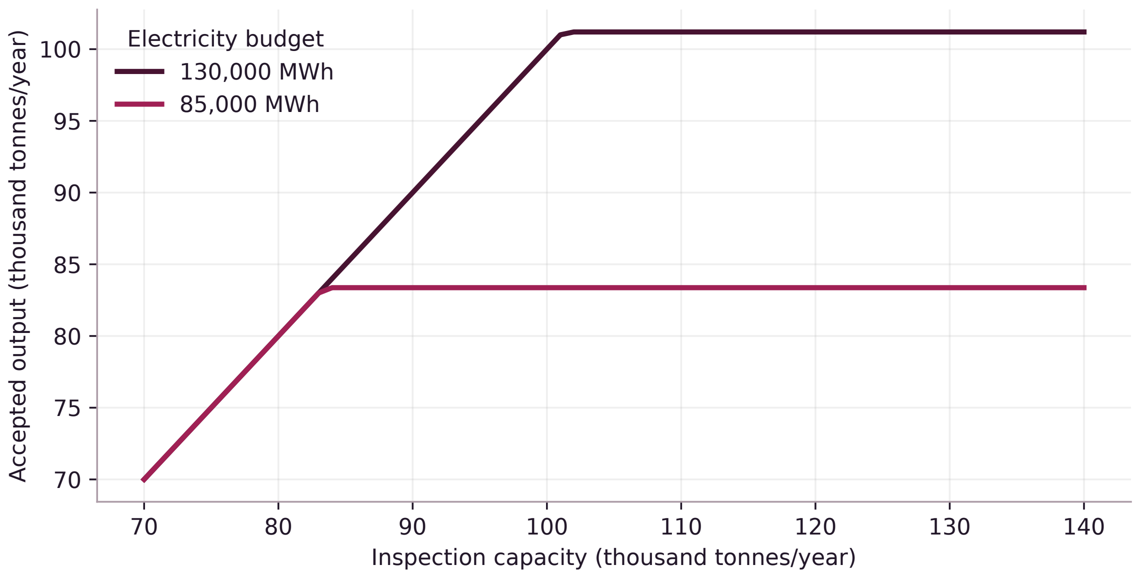

Figure 5: Publication-side throughput frontier. Inspection expansion first helps, then another physical constraint binds. This is not an operating plant result.

15 Resource duals as a disciplined mission selector

Proposition 9 (A shadow-price upper bound)**.** *Let* $v(b) = \max\{c^T x : Ax \leq b, x \geq 0\}$ *be feasible and bounded. If* $\lambda^\star \geq 0$, $A^T\lambda^\star \geq c$ *and* $b^T\lambda^\star = v(b)$, *then for an expanded resource vector* $b + \Delta b$,

$$v(b + \Delta b) - v(b) \leq (\lambda^\star)^T \Delta b. \tag{17}$$

Proof. The old dual vector remains dual-feasible because only the right-hand side changes. Weak duality gives $v(b + \Delta b) \leq (b + \Delta b)^T \lambda^\star$. Subtract the old equality. An active-set change can make the inequality strict. □

This is a standard duality consequence, specialized to mission selection.[12] A zero shadow price on recovery in the inspection-bound example means that marginal recovery capacity has no first-order benefit under that model. It is not a universal statement that recovered material has no value. A binding constraint's multiplier can be nonunique at a kink; the associated finite expansion must be recomputed rather than extrapolated indefinitely.

The executable supplement records primal residuals, dual feasibility, complementarity and objective gaps. The objective only maximizes accepted tonnes; where fabrication input is not uniquely determined, the solver may choose any optimal input. It is not claimed to minimize operating cost or work-in-progress.

16 From material recovery to the thermodynamics of separation

The industrial horizon is constrained by more than mass conservation. Separation requires a change in physical free energy. For an isothermal, isobaric, ideal binary mixture containing n moles with recoverable mole fraction x, complete reversible separation into pure products has minimum work

$$W_{\text{min}} = -nRT\,[x \ln x + (1-x)\ln(1-x)]. \tag{18}$$

It follows from the ideal-mixture Gibbs free-energy difference between separated and mixed states. Per mole of recovered species, $w_{\text{min}} = RTH_2(x)/x$; as $x \to 0$, $w_{\text{min}} \sim RT(1 - \ln x)$. Concentration matters even before kinetics and machinery are introduced.

Proposition 10 (Separation is not free recycling)**.** *At fixed $T > 0$ and $0 < x < 1$, the ideal complete-separation work (18) is positive. Real dissipative operation cannot beat this reversible free-energy bound while retaining the same thermodynamic states and environmental assumptions.*

Proof. Both logarithms are negative, so the expression is positive. The second law requires nonnegative total entropy generation; for isothermal-isobaric reversible work the free-energy increase is the minimum, with irreversible dissipation adding to the requirement. The claim does not apply unchanged to different product purities, reactions or access to additional free-energy reservoirs. □

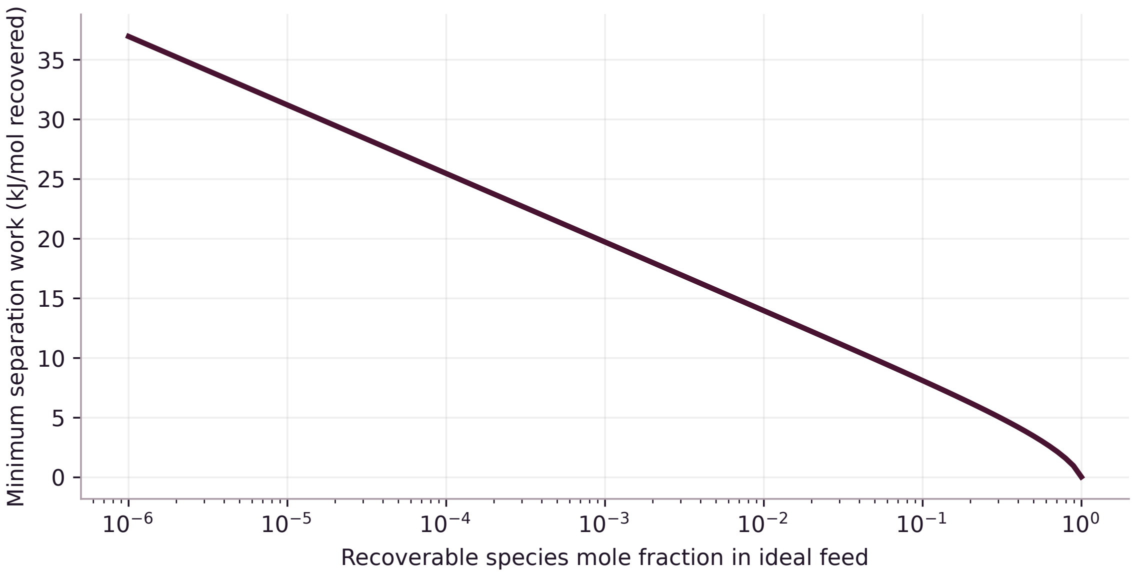

Figure 6: Ideal binary complete separation at 300 K, expressed per mole recovered. This is a reversible thermodynamic bound, not the specific energy of the synthetic recovery plant or a proposed recycling technology.

A real recovery mission must add composition, chemical potential, impurity specifications, process temperatures, reaction pathways, transport, kinetics and equipment life. A learned mass-fraction function does not supply those missing state variables. The distinction between thermodynamic work, information and monetary value must remain explicit.[13]

17 Energy networks, dissipation and thermal admissibility

A physical installation may be represented, when appropriate, by a port-Hamiltonian model

$$\dot{x} = (J(x) - R(x))\nabla H(x) + G(x)u + d, \quad J = -J^T, \quad R \succeq 0, \quad y = G^T \nabla H. \tag{19}$$

Here H is physical stored energy; unlike an institutional score, it has joule units.[14]

Proposition 11 (Power balance and disturbance burden)**.** *For differentiable H,*

$$\dot{H} = -\nabla H^T R \nabla H + y^T u + \nabla H^T d. \tag{20}$$

If $d^T W^{-1} d \le 1$ with $W \succ 0$, the disturbance contribution is at most $\sqrt{\nabla H^T W \nabla H}$.

Proof. Differentiate H, substitute the dynamics and use skew symmetry to eliminate the J term. The disturbance bound is Cauchy–Schwarz after inserting $W^{1/2}W^{-1/2}$. □

For a lumped thermal store, $C\dot{T} = P_{\text{in}} - P_{\text{out}} + w$. Maintaining $T \le T_{\max}$ requires an available control satisfying $P_{\text{out}} \ge P_{\text{in}} + w_{\max}$ on the boundary, under the declared disturbance set and regularity assumptions. If no actuator can deliver that heat rejection, a controller cannot make the state safe by selecting a different label. No such equipment controller is implemented by the present study.

For a physically specified Markov jump process with local detailed balance, entropy production involves state probabilities and transition-rate ratios. A common stationary expression is

$$\dot{S}_i = \frac{k_B}{2} \sum_{i,j} (p_i k_{ij} - p_j k_{ji}) \ln \frac{p_i k_{ij}}{p_j k_{ji}} \ge 0. \tag{21}$$

The inequality follows from $(a-b)\ln(a/b) \ge 0$. Physical interpretation requires a thermodynamic embedding of rates. The existence of a software Markov chain alone does not turn its score into entropy production.[13]

18 An expanding estate must replace itself

Let $m(a,t)$ denote installed mass density of age a. With age-dependent retirement hazard $\mu(a)$,

$$\partial_t m + \partial_a m = -\mu(a)m, \qquad m(0,t) = B(t). \tag{22}$$

Writing survival as $S(a) = \exp[-\int_0^a \mu(s)ds]$, balanced exponential installation $B(t) = B_0 e^{gt}$ gives $M(t) = B(t)\int_0^\infty e^{-ga} S(a)\, da$. This is an ideal age-structured account; a real initial stock also has a transient age distribution.

Proposition 12 (Accepted installations, feed and virgin input)**.** *For deterministic lifetime L, growth $g > 0$, accepted installation mass B, manufacturing yield η and immediately available recovered fraction r of retirements,*

$$B/M = \frac{g}{1 - e^{-gL}}, \qquad R_{\text{ret}} = Be^{-gL}, \tag{23}$$

$$F = B/\eta, \qquad V_{\text{virgin}} = B(\eta^{-1} - re^{-gL}). \tag{24}$$

At $g = 0$, $B = M/L$. At $\eta = r = 1$, positive growth still requires $V_{\text{virgin}} = gM$.

Proof. Integrate survival $S(a) = 1$ for $0 \leq a < L$ and zero otherwise. Retirement equals installations L years earlier. Divide accepted installation mass by yield to get feed, and subtract recovered retirements. The zero-growth limit follows by expansion of e^{-gL}. With perfect yield and recycling, $B - R_{\mathrm{ret}} = gM$. □

At $M = 10^{22}$ kg, $g = 0.02$ per year, $L = 50$ years, $\eta = 0.9$ and $r = 0.85$, gross feed is 3.516×10^{20} kg/year and virgin input remains 2.526×10^{20} kg/year. This is not financed or achieved by the twelve-year project. Recycling delays, losses, composition and repair of the recycling infrastructure would add constraints.

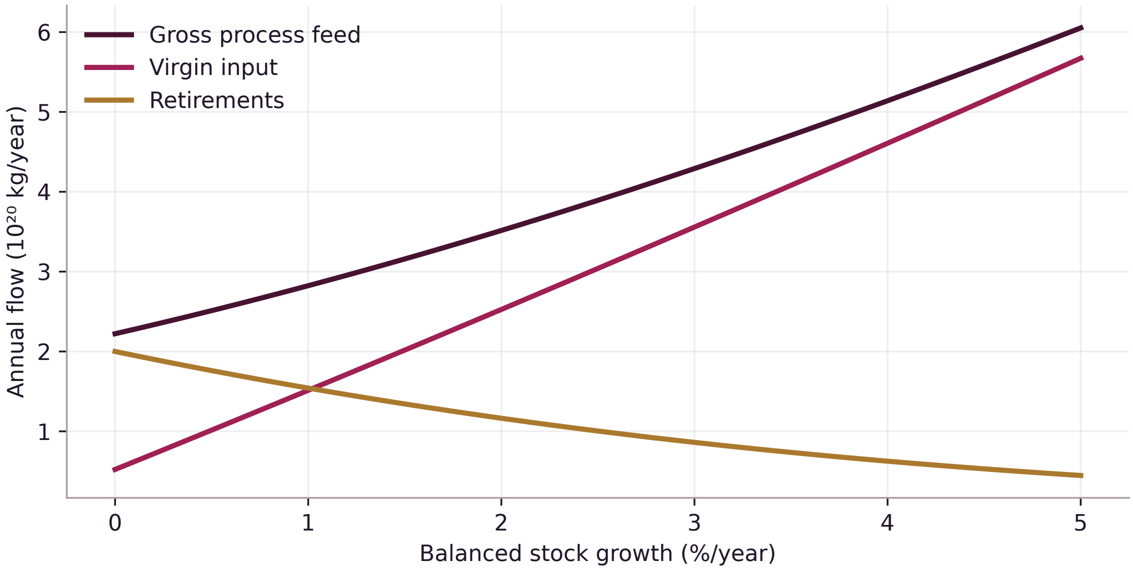

Figure 7: A separately authored balanced-growth resource account. Retained stock, accepted installations, process feed and recovered retirements are distinct quantities.

19 The stellar horizon: power and heat

Let useful power be P_u, conversion efficiency η_c and service availability a. Then required captured power is $P_c = P_u/(\eta_c a)$. For $P_u = 10^{26}$ W, $\eta_c = 0.3$ and $a = 0.97$, $P_c = 3.436 \times 10^{26}$ W, or 89.77% of the IAU nominal solar luminosity 3.828×10^{26} W. The nominal reference irradiance at one astronomical unit is 1361 W/m^2.[16]

Under ideal, unshadowed interception at that flux, collector area is 2.525×10^{23} m^2. If all captured energy eventually thermalizes inside the estate, an effective 600 K, emissivity-0.9 radiator against a 3 K sink needs

$$A_r = \frac{P_c}{\epsilon\sigma(T^4 - T_0^4)} = 5.196 \times 10^{22}\ \mathrm{m}^2. \tag{25}$$

At 20% conversion efficiency, the same target exceeds one nominal star's power. These screens omit geometry, shadowing, structures, delivery losses, construction energy and correlated outages.

For ideal isotropic blackbody radiation, exergy density relative to an environment at T_0 is obtained from $(u - u_0) - T_0(s - s_0)$, where $u = a_R T^4$ and $s = 4a_R T^3/3$. The resulting factor is

$$\psi = 1 - \frac{4}{3}\frac{T_0}{T} + \frac{1}{3}\left(\frac{T_0}{T}\right)^4. \tag{26}$$

It is a blackbody-radiation exergy relation, not an assertion that a real diluted, spectrally selective collector converts sunlight at that efficiency.[15]

IV

Own the next useful decision

A practical path from evidence to a retained capability.

CONCEPTUAL ART · NOT A MEASUREMENT OR DEPLOYMENT

20 The conditional capital-to-capacity pathway

In the favourable recovery state, the separate continuation pays CA$ 360 million for equipment, the CA$ 6.5 million assay ceiling and the CA$ 12 million programme allowance: CA$ 378.5 million at time zero. No operating benefit arrives for two commissioning years. Annual net then starts at CA$ 107.026 million, with a separately authored 1% annual reduction. Initial capital is returned before surplus is allocated.[4]

Over twelve modeled years, project NPV after these expenditures is CA$ 213.945 million; contributed capital of CA$ 378.5 million is returned; CA$ 128.972 million goes to science, the same amount to surplus distributions, and CA$ 386.917 million remains for replacement. Returned capital is not profit. Under the unsuitable state, the same spending convention produces CA$ −152.862 million NPV. Taxes, financing and omitted costs remain outside the authored account.

The Navigator's ladder—identified, modeled, validated, authorized, implemented, realized, collected or conservatively attributable, reinvestable—prevents an estimated value from masquerading as spendable cash.[1] A scenario can examine a successful branch without representing it as the actual output of the current research institution.

The most attractive plausible development route is therefore concrete: retain recovery understanding; qualify measurements; use the best suitable predictor; establish an independently supported operating difference; return contributed capital; then fund the actual next bottleneck. As constraints move, the next specialist may concern inspection, heat, fabrication, logistics or repair rather than a larger version of the preceding model.

21 How to use SUCCESSOR Ω: the owner route

21.1 Choose one decision with an observable outcome

Begin with a decision repeated often enough to learn from and consequential enough to justify evaluation. Name the accountable owner, the population, the measurement, the time horizon and a credible alternative. Examples include a bounded invoice review, a maintenance schedule or an experimental-design choice. Do not start by granting enterprise-wide autonomy.

21.2 Constitute before training

Write a one-page mission: what decision is proposed, what counts as success, what must never happen, what evidence is permitted, what comparison will be fair, what complete costs matter and what observation would stop the programme. Free text is a draft until a supported adapter expresses its operative parts. The general workspace included in this package produces a handoff brief, not a compiled authority policy.

21.3 Separate evidence by role and preserve exposure

Keep training, development selection and challenge evidence distinguishable. Record provenance, rights, event identities, transformations and known prior use. A new filename does not make an old observation fresh. Synthetic examples teach a workflow; they do not silently replace missing customer evidence. Do not insert confidential evidence into a local plaintext workspace before its storage and rights have been reviewed.

21.4 Compare the complete alternatives

Reproduce the incumbent and a serious alternative under the same permitted observations and budgets. Record local proxies as local proxies. Include formation, execution, human rescue, maintenance, dependence and unwind. A useful negative result is retained; no stage should automatically produce a success designation.

21.5 Freeze, evaluate, decide, retain

A useful candidate has an exact model, preprocessing, programs, protocol and evidence identity. Evaluate that composition, preserve failures, and decide whether to retain the parent, keep the child for further research, substitute Beta or stop. An export must be restorable and subject to its rights boundary. A local signature establishes attribution only to the relevant key; it is not independent capability proof.

A good first session ends with a better decision

Leave with one mission, one owner, one fair baseline, explicit evidence gaps, a bounded next experiment and a named review. It is a success to discover that the wrong problem was being automated.

22 How to use the delivered research package

22.1 No-code inspection

Open `START_HERE.html`. Read the owner brief, then open the retained ANABASIS Mission Command. Follow both positive and negative branches. Change the assumed assay bias or cost and inspect why the policy changes. The new Evidence Boundary laboratory shows frozen-policy channel sensitivity and computes simple physical and incentive examples. It does not retrain the native models.

22.2 Reproduce the historical v10.2-based case

From `case_study/ANABASIS_OMEGA_1_0`, using the separately prepared, documented Python environment:

```
python3 VERIFY_PACKAGE.py
python3 SUCCESSOR.py doctor
python3 REPRODUCE.py --out var/replay_001
python3 -m pytest tests -q
```

Use a new output directory. The retained replay uses NumPy 2.3.5 for its numerical lineage; dependencies are not bundled. It runs local research, not a provider service or equipment controller. The root `REPRODUCE_PUBLICATION.py` runs the new extension and directs its outputs into a fresh directory; it does not overwrite the historical trial.

22.3 The separately delivered v11 workflow

The newer v11 product is a distinct implementation release. Its retained guides are included for current general use, but the v11 product ZIP is not duplicated here and ANABASIS has not been retrospectively reclassified as a v11 experiment. From an installed/extracted matching v11 product, its documented route is:

```
python3 SUCCESSOR.py doctor
python3 SUCCESSOR.py init --workspace var/my_mission
python3 SUCCESSOR.py serve --workspace var/my_mission
```

Open the private loopback session link printed by the server. Validate and confirm the exact plan, train and freeze, supply the challenge, inspect actual comparators, and create a separately evaluated child. Initialize once; later sessions use `serve`. Its supported adapter has at most four numerical inputs and one numerical target. A natural-language description does not automatically become an arbitrary business process.

Use the v11 guides only with their matching source, environment and release contract. This monograph does not claim a fresh full v11 regression qualification; the fresh execution in this edition concerns ANABASIS and its publication-side analyses.

23 From trained components to an earned specialist

The next real deployment would need instrument qualification, representative observations, protected evidence custody, independent comparison, source rights, current full economics, target-environment security and recovery, and a named policy authority. The three source-defined objects remain distinct throughout.[1, 2, 3]

The 21-job source map records what is available and what is missing. In particular, independent validation, protected Fresh-Proof evaluation and authority admission are not emulated by fabricated actors. A local evidence file is a request or internal result, not a certificate from an institution that did not participate.

The recursion diagram on PDF page 48 of the Gym paper explicitly routes a successor through descendant formation, fresh proof and accountable admission. A later generation may inherit useful context and admitted capabilities under rights; it does not inherit proof or operating power. The attached framework's stronger recursive-improvement claim additionally requires better formation or governance on fresh work without weaker proof. The present extension studies mechanisms relevant to that goal; it does not claim to have completed that institutional demonstration.

Light takes about 998 seconds for a one-AU round trip. Subsecond thermal obligations therefore require qualified local control, not a remote unrestricted agent. These physical obligations establish neither infrastructure nor an arrival date.

24 Limitations, validation and the next scientific question

24.1 What has been checked

This publication freshly runs the retained 68-test suite and replays the four-cohort study. It also executes new tests of the posterior sensitivity, fixed-policy loss boundary, resource primal/dual witnesses, material conservation, Brier incentive identity, audit feasibility, physical power limits and Gibbs identity. Exact counts, actual environment, replay comparisons and clean-package integrity results are written to the accompanying validation record rather than assumed from earlier editions.

24.2 What remains uncertain

The central scenario was selected to reveal a mechanism, not sampled from a population of projects. Hidden states are few; kernels and financial outcomes are authored; process evidence is synthetic; observation adequacy is not established by the likelihood table; the informed comparator is not a market survey; the original severe-loss definition omits assay and programme costs; the holdout is local, not independently protected. Mathematical results inherit their hypotheses. No diagram or high-resolution illustration removes these boundaries.

24.3 A concrete research agenda

First, measure the observation channel that most strongly influences the capital decision, including joint failures. Second, compare complete research strategies under the same evidence rights and costs. Third, calibrate transfer on a newly constituted mission and disclose every adaptation. Fourth, test an independent export-and-rebuild with a replaceable supplier. Only then can a customer determine whether the owned institution earns advantage after the relevant cost and basis-risk reserves.

The expansion institution

Intelligence becomes useful infrastructure only through accountable transformations: a hypothesis becomes a tested program; a program changes a decision; an accepted decision creates qualified output; collected surplus funds the next instrument and capability. The institution owns that path and remains capable of refusing its own preferred answer.

A Notation and units

Symbol	Meaning and scope
$w = (r, f, b)$	Hidden recovery suitability, fabrication suitability and persistent assay bias.
h, p_h, L	Evidence history, posterior belief and observation kernel.
U, V, c_e	Project NPV before assays; continuation value net of future assays; assay fee. Millions of Canadian dollars in the finite case.
ρ, λ	Frozen loss-probability ceiling; interpolation between two authored channels. Distinct concepts.
M, B, R_{ret}, F	Installed stock; annual accepted installations; annual retirements; annual process feed.
g, L, η, r	Stock growth, deterministic life, manufacturing yield, recycling fraction; not parameters of the original financial trial.
P_u, P_c, a	Useful power, captured power, service availability.
H, J, R, G	Physical stored energy, skew interconnection, positive dissipation and input map in a proposed physical model.
τ	Computational score temperature in the Gibbs identity; not kelvin.

B Source-to-artifact map

Source basis	Implementation or publication location
Navigator v5.0.0, PDF pp. 31–34	SEIZE sequence, separate five decisions, frozen risk and cost definitions; mission constitution and handoff.
Gym v2.1.0, PDF pp. 16–21	Hidden state, observations, bounded actions, role and maturity distinctions; retained `MissionGym`.
Gym v2.1.0, PDF pp. 46–48	Gym revision, Negative Capability Graph, non-inheritance; Sections 6 and 23.
Gym v2.1.0, PDF pp. 73–75	Twenty-one bounded jobs; retained `protocol/BOUNDED_JOBS.json`.
Corporate white paper, pp. 19–21	Neural-symbolic cognition and separate serving/learning branches; local research boundary.
Corporate white paper, pp. 30–37	Alpha, authority and portability; Sections 7, 20–23.
ANABASIS simulation 1.0	Unchanged `case_study/`; learned artifacts, policies, observations and physical/economic accounts.
This publication	`research/extension.py`, results, tests and numerical figures; no change to the original trial.

C A reusable one-page mission constitution

Decision and owner. What exact repeated decision matters, for whom, and who can refuse it?

World and evidence. What can be observed at decision time? Which sources are authorized? Which latent states or shared errors could explain agreement?

Alternatives. What do the incumbent and strongest credible available alternative do under the same constraints? Is no-action retained?

Utility and hard gates. Define units, horizon, complete costs, unacceptable events and the loss denominator before seeing results.

Experiment. Name the smallest observation able to change the decision, its full cost, stop condition, data role and invalidation trigger.

Freeze and proof. Bind candidate, source, likelihood, utility, evaluator, adaptation budget and rights. Name actual independent custody where required; leave it absent when it does not exist.

Authority and recovery. State what is permitted now, who can change it, when it expires and how the last supported state is restored.

Value and succession. Keep modeled, realized and reinvestable value separate. Name the next trigger that would reopen the mission.

D Reproduction, provenance and editorial boundaries

The deliverable contains editable LaTeX, a bibliography data file, deterministic numerical scripts, retained case evidence, the exact source PDFs inside the case package, and the new validation record. Build scripts use XeLaTeX with system fonts; no font files are redistributed. The core trial retains its v10.2 lineage. v11 manuals are reference companions, not hidden migrations of the experiment.

The standard build creates English and French papers, the bilingual collected edition and a practical guide. The cover and part plates use AI-generated art cropped to remove non-authoritative charts where possible. The full supplied conceptual plate is retained as artwork, not as a technical results page. Original images, crop coordinates and hashes are recorded. Numerical plots use actual result files.

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[3] MONTREAL.AI. *Neural-Symbolic SUCCESSOR Ω: Customer-Owned Mission Intelligence*. Corporate white paper, 25 August 2026. Supplied 59-page architecture; retained case copy: `docs/Architecture_Reference_EN.pdf`.

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